

ALAN HUYNH

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SKILLS

Data Science & ML: PyTorch, TensorFlow, Scikit-Learn, Gen AI, Time-Series, Statistical Models, Computer Vision
Languages/Libs: Python, SQL, JS/TS, C/C++, Java; NumPy, Pandas, SciPy, PySpark, Rasterio, GeoPandas
DevSecMLOps: AWS, Azure, GCP, Docker, Terraform, Pulumi, GitHub Actions, CI/CD, Linux, Streamlit

EXPERIENCE

Tech Lead

Sep. 2025 – Present

ProtoFlow Pty Ltd

Melbourne, VIC

- Led a cross-functional team of 3 engineers delivering an AI-powered multilingual subtitling platform, transcribing and translating 1,000+ hours of video content across up to 10 languages.
- Architected full DevSecOps lifecycle with security-first design: Terraform-provisioned AWS infrastructure, GitHub Actions CI/CD pipelines, database migrations, and telemetry; supporting 20+ active enterprise users with zero unplanned downtime.
- Led agile delivery across a cross-functional team of 3, running sprint planning and stakeholder communication end-to-end for a platform processing 3,000+ subtitle edits per day; enforced quality through unit, integration, and load testing alongside structured code reviews.
- Built a multi-agent MQM translation quality pipeline orchestrating Claude, Gemini, and GPT; integrated Deepgram for automated speech transcription, subtitle alignment, and context-aware editing; deployed a Moodle LLM chatbot supporting 200+ students daily.

Machine Learning Engineer

Dec. 2024 – Mar. 2025

Fair Dinkum Systems

Melbourne, VIC

- Engineered dual-stream Vision Transformer architectures in PyTorch, boosting battery RUL prediction stability and accuracy by 30% across 6 datasets, directly informing grid storage decisions for energy clients.
- Developed custom time-series pipeline integrating wavelet transforms, spectrograms, and Fourier analysis for enhanced signal representation.
- Implemented contrastive learning framework improving model robustness and out-of-domain generalization by 25%; collaborated within a 4-person team, maintained documentation in Notion, and ensured quality through pytest and GitHub Actions.

Software Engineer

Sep. 2024 – Sep. 2025

Math Song Pty Ltd

Melbourne, VIC

- Led full-stack Django platform development, automating operations and enhancing business efficiency.
- Maintained 99.99% uptime while processing \$200K+ monthly through integrated payment APIs.
- Orchestrated GCP infrastructure with Docker, Terraform, CI/CD pipelines, and VM provisioning for scalable deployment.

Machine Learning Engineer

Feb. 2024 – Sep. 2024

A.I.gorithm

Melbourne, VIC

- Delivered ML solutions for Australia's carbon initiative, partnering with major banks and national retailers to monitor emissions and environmental metrics across 300,000+ meshblocks nationwide.
- Built semi-supervised computer vision system for multi-spectral satellite imagery analysis using self-supervised representation learning, preserving 87% accuracy when adapting European frameworks to Australian conditions.
- Developed production ML pipelines combining multi-spectral imagery ETL and temporal feature extraction using AWS Glue (Spark), EMR (Hadoop), and Lambda.

Data Scientist

Jan. 2022 – Jan. 2024

LUX Aerobot

Melbourne, VIC | Hybrid

- Achieved 92% wildfire prediction accuracy using computer vision with multi-spectral aerial imagery, supporting Canadian and Australian environmental monitoring initiatives.
- Optimized ML models for NVIDIA Jetson edge deployment through quantization and TensorRT, reducing model size by 60% and enabling real-time field inference.
- Built geospatial ML pipeline with AWS SageMaker, Neo, and Greengrass, reducing inference latency by 75%.

CERTIFICATION

AWS Certified Machine Learning – Specialty (MLS-C01)

Mar. 2025

PROJECTS

Mornington Peninsula Digital Twin

Feb. 2025 – Present

Python, Sentinel-2, Streamlit, GeoPandas, Rasterio, Plotly

- Built a full-stack geospatial intelligence platform from raw satellite ingestion to a 3D interactive digital twin; automated a Sentinel-2 remote sensing pipeline ingesting 51 scenes (2023–2026) from Digital Earth Australia at 20m resolution.
- Computed 10 spectral indices (NDVI, NDMI, NBR, NDBI, MNDWI, albedo, turbidity, chlorophyll proxy) with cloud masking, atmospheric correction, and seasonal aggregation across 3,252 ABS mesh blocks.
- Delivered a Streamlit web app with 13 switchable surface layers (vegetation health, fire risk, urban heat, coastal turbidity) and 5 demographic choropleth variables; published curated dataset to Kaggle.

Microplastics Estimation

Feb. 2025 – Present

Python, PyTorch, AWS

- Engineered a Physics-Informed Multi-Head CNN for global microplastic estimation, achieving 92% accuracy and RMSE <10%; paper forthcoming.
- Boosted model generalization by 30% on unseen data through custom physics-constrained loss functions; validated via ablation studies against FNOs and GNNs.
- Managed end-to-end ML workflow on AWS EC2 using PyTorch Lightning for reproducible experimentation and hyperparameter optimization.

Torch Activations

Apr. 2023 – Present

Python, PyTorch, Poetry

- Published open-source PyPI library implementing 300+ PyTorch activation functions; 100% test coverage, automated CI/CD via GitHub Actions, full docs on ReadTheDocs.

ACHIEVEMENTS

Green Battery Hack – First Place, Innovation Track

Apr. – May 2024

MLAI Aus

Melbourne, VIC

- Won national hackathon by engineering an RL-based ML model for AEMO spot market battery trading, using custom profit-maximizing loss functions and Optuna-tuned Q-Learning strategies.

Public Speaker – Machine Learning & Artificial Intelligence

DDD Melbourne, AICamp Melbourne

- Presented applied ML/AI concepts to 100+ industry professionals at DDD Melbourne and AICamp events.

Mentor

Feb. 2025

CISSA | University of Melbourne

Melbourne, VIC

Arcaea Melbourne – Community Organiser

May 2025 – Present

University of Melbourne

- Founded and organise monthly rhythm game community meetups at UniMelb, growing a recurring player base from scratch.

EDUCATION

Melbourne Polytechnic

Feb. 2024 – Present

Bachelor of Information Technology

Melbourne, VIC

Deakin University

Feb. 2023 – Oct. 2023

Accelerated Program

Melbourne, VIC